SIAGA Model Card

Version: v1

Task: predict a country's life expectancy at birth (years) from health-system and contextual indicators.

Model: monotonicity-constrained histogram gradient-boosting regressor. **Training rows:** 98 country-years (10 ASEAN states, 2004-2014).

Intended use

Decision support for public-health resource allocation: ranking drivers of life expectancy and simulating the effect of policy changes. Not a clinical or individual-level tool. Operates on aggregate national indicators only.

Performance (held-out)

Test	R2	RMSE (yrs)	MAE (yrs)
Unseen country (leave-one-country-out)	0.686	3.09	2.58
Future of known countries (2012-2014)	0.923	1.46	1.12
Out-of-source vs World Bank (2015-2019)	0.693	2.73	2.29

Features

- `physicians_per_1000` (actionable, monotonic +)
- `nurses_midwives_per_1000` (actionable, monotonic +)
- `pharma_workers_per_1000` (actionable, monotonic +)
- `immunization_dpt` (actionable, monotonic +)
- `immunization_measles` (actionable, monotonic +)
- `tb_prevalence` (actionable, monotonic -)
- `hiv_prevalence` (actionable, monotonic -)
- `undernourished_pct` (actionable, monotonic -)
- `crude_birth_rate` (context, monotonic 0)
- `log_capex_per_capita` (actionable, monotonic +)
- `log_gdp_per_capita` (context, monotonic +)
- `sanitation_pct` (actionable, monotonic +)
- `year` (context, monotonic 0)

Monotonic constraints guarantee the medically correct direction of each effect, so the policy simulator cannot produce a perverse recommendation.

Limitations

- Small panel (10 countries). Generalization to an unseen country is the weakest test; report ranges, not point certainty.
- Mortality indicators are excluded to prevent target leakage.
- The linear surrogate served by the API is an approximation for edge inference.

Provenance

Trained by `python -m pipeline.train`. Source data: ASEAN Statistical Yearbook files and the World Bank Open Data API. See `data_dictionary` for full schema.